

# Piecewise-Affine Local Volatility

Calibrating a Dupire surface from sparse quotes: model, PDE, gradients, and speed

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## Abstract

This note documents the Vol-Fitter’s full local-volatility model: a continuous piecewise-affine ( $P_1$  finite-element) local-variance surface  $\nu_\theta(t, x)$ , calibrated directly to sparse vanilla and variance-swap quotes through the forward Dupire PDE as an implicit pricing map. Unlike the slice models (Notes 01–03) this is a genuine two-dimensional local-volatility surface: strict positivity of the nodal variances makes the implied prices butterfly- and calendar-clean up to scheme noise, which the model *measures* rather than assumes. We derive the normalized forward Dupire equation, the  $P_1$  parametrization and its positivity guarantee, the implicit-Euler discretization and observation operator, the variance-swap log-contract replication and its faster source-PDE alternative, and the calibration objective with its roughness, convex-wing and front-tie regularizers. We then give the sensitivity PDE and the adjoint gradient, and — the engineering heart of the note — the numerical optimizations that cut the cold fit  $\sim 3\text{--}6\times$  (from  $\sim 86\text{ s}$  to  $\sim 15\text{--}30\text{ s}$ , scaling with grid size), with recalibrations near-instant: a compiled vectorized-Thomas Dupire march ( $\sim 6.5\times$ ), a stall-based early-stop, a parametric Dupire warm seed, and the matrix-free Gauss–Newton solver that avoids the dense SVD. A synthetic round trip recovers a known surface to 0.6 vol bps, and the production fit on a static Bloomberg fixture reaches 2.8 vol bps on SPY and 11.7 on NVDA. The note closes with the short-dated-fit fixes and the convex-wing regression that the benchmark caught.

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# 1 The idea

A slice model fits one expiry at a time and stitches expiries together with a calendar constraint. A *local-volatility* model does the opposite: it posits a single function  $\sigma_{\text{loc}}(t, S)$  such that, under  $dS_t = \sigma_{\text{loc}}(t, S_t) S_t dW_t$  (in forward-normalized units), the model reproduces *all* expiries and strikes at once. Dupire showed such a function exists and is unique given an arbitrage-free price surface. The Vol-Fitter follows the Andreasen–Huge philosophy: rather than *differentiate* a fitted price surface to extract  $\sigma_{\text{loc}}$  (numerically explosive in sparse data), it *parametrizes*  $\sigma_{\text{loc}}$  directly and prices *forward* through the Dupire PDE, calibrating the parameters to quotes.

## Heuristic

Extracting local vol by the Dupire formula  $\sigma_{\text{loc}}^2 = \partial_T C / (\frac{1}{2} K^2 \partial_{KK} C)$  divides two noisy numerical derivatives of a sparse surface — a recipe for garbage in the wings. Calibrating local vol *forward* instead never differentiates the data: it asks “what positive surface, run through the PDE, best reprices the quotes?” Positivity of the surface is imposed on the *parameters*, so every iterate is arbitrage-clean by construction — the same philosophy as LQD, one dimension up.

## 2 The forward Dupire equation

Work in forward-normalized strike  $x = K/F_T$  and let  $c(T, x)$  be the normalized call. Writing local variance  $\nu(t, x) = \sigma_{\text{loc}}^2$ , the forward Dupire equation is

$$\partial_T c(T, x) = \frac{1}{2} \nu(T, x) x^2 \partial_{xx} c(T, x), \quad c(0, x) = (1 - x)^+. \quad (1)$$

It is a forward parabolic PDE in  $(T, x)$  with the call payoff as initial data. Two facts make it the right object. First,  $\partial_{xx} c \geq 0$  is exactly the butterfly (density) condition, and  $\nu \geq 0$  keeps the equation parabolic, so a positive  $\nu$  *propagates* non-negativity of the density forward in maturity. Second, monotonicity of  $c$  in  $T$  (the calendar condition) follows from (1) whenever  $\nu \geq 0$ , since the right-hand side has the sign of  $\partial_{xx} c \geq 0$ . Thus a strictly positive local-variance surface is *structurally* free of both static arbitrages, up to the discretization noise the model measures (section 4.2).

## 3 The piecewise-affine surface

### 3.1 Parametrization

The local variance is a continuous piecewise-affine ( $P_1$ ) function on a tensor grid of  $n_t \times n_x$  vertices  $(\tau_i, \xi_j)$ ,

$$\nu_\theta(t, x) = \sum_{i,j} \theta_{ij} \phi_{ij}(t, x), \quad (2)$$

where  $\phi_{ij}$  are the bilinear (or barycentric/Delaunay) hat functions and  $\theta_{ij} = \nu(\tau_i, \xi_j) \geq 0$  are the *nodal local variances* — the calibration parameters. The flat vector is  $\theta = \text{vec}(\theta)$ , time-major.

**Proposition 1** (Nodal positivity  $\Rightarrow$  surface positivity). *A  $P_1$  function is a convex combination of its vertex values on each cell. Hence if  $\theta_{ij} \geq 0$  for all  $i, j$ , then  $\nu_\theta(t, x) \geq 0$  everywhere, and  $\nu_\theta > 0$*

if the nodal values are bounded below by a positive constant.

This is why the calibration enforces only *box bounds* on  $\theta$  (default local-variance bounds  $[0.005, 0.20]$ , i.e. local vol in  $[7\%, 45\%]$ ): positivity of the whole surface reduces to positivity of finitely many numbers. Beyond the lowest strike vertex the surface extrapolates the local variance *linearly* with a slope set by the first interior cell (`left_extrap_a`; Note 09), rising toward  $x \rightarrow 0$ ; the right wing is flat-clamped.

## 3.2 Grid design

The vertex grid is delta-spaced in strike and floored per expiry so that even a narrow short-dated smile lands enough vertices on its curvature (`gridXNodes`, `gridXMinPerExpiry`; section 9). The time vertices (`gridTNodes`) carry the term structure. The grid is deliberately modest — a dozen strike vertices, ten in time — because each vertex is a free parameter and the PDE solve cost scales with the grid; the regularizer (section 6) supplies the smoothness that a coarse grid would otherwise lack.

# 4 The pricing map

## 4.1 Discretization

The forward map (1) is solved on a uniform spatial grid with step  $\Delta x = 0.01$  on  $x \in [0, \sim 2.5]$  ( $\sim 250$  nodes) by fully implicit (backward) Euler in maturity with step  $\Delta T = 0.01$  (the local variance entering each step at the new time level  $t_{n+1}$ ), optionally Rannacher-started with a couple of implicit start-up steps before Crank–Nicolson. (A separate log-strike validator, appendix A, runs the finer fixed grid.) Each step solves a tridiagonal  $M$ -matrix system

$$(I - \Delta T A_n) U^{n+1} = U^n, \quad (3)$$

where  $A_n$  encodes  $\frac{1}{2}\nu x^2 \partial_{xx}$  at step  $n$ . The matrix is an  $M$ -matrix (diagonally dominant, non-positive off-diagonals), so the scheme is unconditionally stable and monotone — it cannot create spurious negative densities. The observation operator reads the normalized call at each quote’s  $(T, x)$  off the solution by interpolation.

Diagonal dominance makes pivoting unnecessary, so each step (3) is a no-pivot Thomas march — the kernel the compiled march (appendix B) accelerates  $\sim 6.5\times$ :

Listing 1: The no-pivot tridiagonal (Thomas) solve for one implicit step (3) (distilled from `models/localvol/affine_march.py`; matches a dense solve to  $10^{-10}$ ).

```

1  def thomas(a, b, c, d):                # a: sub-diagonal, b: diagonal, c:
    super-diagonal
2      n = len(d); cp = np.empty(n); dp = np.empty(n)
3      cp[0], dp[0] = c[0]/b[0], d[0]/b[0]
4      for i in range(1, n):              # forward sweep: eliminate the sub-
        diagonal
5          m = b[i] - a[i]*cp[i-1]
6          cp[i], dp[i] = c[i]/m, (d[i] - a[i]*dp[i-1])/m
7      x = np.empty(n); x[-1] = dp[-1]
8      for i in range(n-2, -1, -1):      # back substitution

```

```

9         x[i] = dp[i] - cp[i]*x[i+1]
10    return x

```

## 4.2 No-arbitrage diagnostics

Because the surface is only *up to scheme noise* arbitrage-free, the model measures the residual: a butterfly tolerance on  $\partial_{xx}c$  ( $\gtrsim -10^{-6}$  density) and a calendar tolerance on  $\partial_T c$  ( $< -10^{-9}$  price decrement). The model is gated behind these diagnostics rather than assuming cleanliness — a deliberate design choice.

## 5 Variance swaps

A variance-swap quote is the fair strike of realized variance, which for a diffusion equals the expected total variance to  $T$ . Two pricers are available. The *static* method uses the log-contract replication

$$I(T) = 2 \int_0^1 \frac{P(T, x)}{x^2} dx + 2 \int_1^\infty \frac{C(T, x)}{x^2} dx, \quad (4)$$

a trapezoid over the priced surface (half-width and node count `VS_HALF_WIDTH`, `VS_POINTS`). The faster *source-PDE* method (default-off, gated by `varSwapMethod`) solves one backward source PDE  $\partial_t g + \frac{1}{2}\nu x^2 \partial_{xx} g + \nu = 0$ ,  $g(T, \cdot) = 0$ , and reads  $I(T) = g(0, 1)$  — with analytic sensitivities  $\partial I / \partial \theta$  from the same multi-RHS machinery, and without the  $x_{\max}$ -dependence of the replication tail. Note 08 develops both in full.

## 6 The calibration objective

The fit minimizes a stacked least-squares residual over  $\theta$ ,

$$\min_{\theta \in [\nu_{lo}, \nu_{hi}]} \frac{1}{2} \left( \underbrace{\|r_{\text{opt}}\|^2}_{\text{quotes}} + \underbrace{\|r_{\text{vs}}\|^2}_{\text{var-swaps}} + \underbrace{\lambda \|L(\theta - \theta_{\text{ref}})\|^2}_{\text{roughness}} + \underbrace{w_{\text{cvx}} \|r_{\text{cvx}}\|^2}_{\text{convex wing}} + \underbrace{w_{\text{ft}} \|r_{\text{ft}}\|^2}_{\text{front tie}} \right), \quad (5)$$

with the blocks:

- **Option residuals**  $r_{\text{opt},q} = (P_q(\theta) - y_q) / \eta_q$  in vega-normalized forward-call price units (mid), or the bid–ask/hairecut band hinge with a small mid anchor when band edges are present (Note 07).
- **Var-swap residuals**  $r_{\text{vs}} = (I(\theta) - z) / \zeta$  (Note 08).
- **Roughness**  $\sqrt{\lambda} L(\theta - \theta_{\text{ref}})$  with  $L$  the spacing-aware second-difference operator and  $\theta_{\text{ref}}$  a reference surface; the time-vs-strike balance is `gridRegRho`. This is the smoothness that lets a coarse grid generalize.
- **Convex wing** hinge: a one-sided penalty enforcing local convexity of the wing, confined to vertices *below the deepest quote* (the extrapolation tail only — see section 9.1).
- **Front tie**: a light penalty tying the unconstrained  $t = 0$  time row to the first (data-identified) expiry row across strikes, anchoring the front end where the shortest expiry has little coverage.

## 7 Sensitivities and the adjoint

### 7.1 The sensitivity PDE

Each parameter  $\theta_\ell$  enters (1) linearly through  $\nu_\theta$ , so the price sensitivity  $s_\ell = \partial c / \partial \theta_\ell$  obeys the same parabolic PDE with a *source*:

$$\partial_T s_\ell = \frac{1}{2} \nu_\theta x^2 \partial_{xx} s_\ell + \frac{1}{2} \phi_\ell x^2 \partial_{xx} c, \quad s_\ell(0, \cdot) = 0. \quad (6)$$

Discretized, every  $s_\ell$  shares the *same* tridiagonal factor as the value solve (3), differing only in the source — so the Jacobian columns are a single multi-RHS back-substitution against one factorization. This is the forward-sensitivity route,  $O(N_t N_x m)$  for  $m$  active parameters.

### 7.2 The adjoint gradient

When only the *gradient* of the scalar objective is needed (not the full Jacobian), the discrete adjoint computes it in one forward value solve plus one backward adjoint sweep,  $O(N_t N_x)$  *independent of  $m$* . The Vol-Fitter uses the forward sensitivities for the Gauss–Newton Jacobian (it needs the columns) and keeps the adjoint as the route for very large vertex counts.

## 8 Solvers and the speed story

The cold fit of (5) was, at  $\sim 533$  vertices, an  $\sim 86$  second affair hitting the 200-evaluation cap — the single heaviest computation in the app. Driving it down is most of the engineering in this model, and the lessons are worth stating because several plausible ideas *failed*.

### 8.1 Where the time goes

On a cold fit the cost splits roughly as: optimizer linear algebra / dense SVD  $\sim 52\%$ , PDE sensitivity march  $\sim 32\%$ , Jacobian assembly  $\sim 14\%$ , value solve  $\sim 2\%$ . No single per-evaluation lever moves the total, because the cost is distributed — which is why four separate “faster per evaluation” attempts (coarse grid, first Numba try, Rannacher, and the first matrix-free GN) each won only  $\sim 1.1\text{--}1.2\times$  and were shelved.

## 8.2 What worked

### Performance

The cold fit is now  $\sim 3\text{--}6\times$  faster than the original banded baseline (scaling with grid size); recalibrations are near-instant via warm start. The levers, in order of impact:

1. **Stall-based early-stop.** Track the option-block misfit and stop at the best iterate once it stalls. Because fewer evaluations multiply march, assembly *and* optimizer together, this scales the *whole* fit: measured  $1.45\times$  (SPY) to  $3.3\times$  (NVDA), at  $+0.1\text{--}0.25$  vol bp.
2. **Compiled vectorized-Thomas march.** A `@njit` no-pivot Thomas solver with the sensitivity columns as the contiguous SIMD inner loop and a fused source:  $\sim 6.5\times$  the LAPACK banded march, numerically exact to  $\sim 10^{-15}$ . The first Numba attempt (column-outer scalar Thomas) won only  $1.2\times$  — the real lever was the loop order.
3. **Matrix-free Gauss–Newton.** The default solver avoids `trf`’s dense SVD (52% of an evaluation) by solving the GN step with preconditioned LSMR on matrix-free  $Jv/J^\top w$  products. Once the march is cheap, the no-SVD evaluations win:  $\sim 1.3\text{--}1.65\times$  over TRF. Gated to the smooth mid-fit target.
4. **Parametric Dupire warm seed.** Seed  $\theta$  from the parametric surface’s local variance ( $\sim 1.3\text{--}1.8\times$  on cold fits) and from the previous surface on recalibration ( $\sim 38\times$ ).

The order-of-magnitude wins are spent; remaining levers are incremental.

### Caution

Removing the dense SVD made fits *slower* at  $\leq 440$  vertices, because there the bottleneck is the PDE march, not the SVD — the SVD wall is a  $\gtrsim 1000$ -vertex (future non-tensor “bowtie” grid) phenomenon. The clean synthetic benchmark (zero residual, in bounds) hid this: GN converged in 8 evaluations there but did *not* converge within the cap on stiff real data. The honest lesson: measure on real chains, and distrust a synthetic-only speed-up.

## 9 Short-dated calibration

A genuine one-week smile fit *catastrophically* (108 vol bp RMS vs the parametric  $\sim 47$ ) while normal expiries fit well. A measure-first diagnosis (per-expiry side-channel counting in-range vertices, vega flooring and PDE steps) ruled out the usual suspects — vega floor, time steps, local-vol cap, prior leakage — and isolated *strike under-resolution*: the delta strike axis is sized to the *longest* expiry and clipped to the global range, so a narrow short smile landed only  $\sim 3$  vertices on its sharpest curvature. Two fixes:

1. **Per-expiry coverage floor** (`gridXMinPerExpiry`, default 8): after the delta axis is built, split the widest in-range gaps until each expiry has at least  $m_{\min}$  vertices inside *its own* traded range — adding nodes only to under-covered short fronts.
2. **Short-expiry-aware PDE step:** refine the shared PDE  $x$ -step to  $0.3\times$  the smallest ATM  $\sigma\sqrt{\tau}$ , snapped so  $x = 1$  stays a node, clamped to  $[1/400, 0.01]$ . Normal surfaces stay on 0.01

(byte-identical).

The one-week fit fell from 108  $\rightarrow$  23.5 vol bp (now *better* than the parametric), with the long expiries byte-identical.

## 9.1 The convex-wing regression

The benchmark also caught a subtler bug. The convex-wing constraint originally selected *every* vertex in the deep wing regardless of data; at a user's saved `gridXNodes = 20` it stacked convexity penalties onto densely-quoted put strikes and forced the *wrong* wing on low-vol SPY (NVDA's naturally convex wing had hidden it). The fix confines `convex_cols` to vertices *below the deepest quote* — the extrapolation tail only. SPY went 25.7  $\rightarrow$  2.6 vol bp. The lesson echoes Note 09: a shape constraint must not be imposed where the data already speaks.

# 10 Worked examples

## 10.1 Synthetic round trip

A known skewed local-variance surface is priced through the forward PDE, quotes are generated at four expiries, and the surface is recovered from a *flat* seed by the production calibrator. Figure 1 shows the recovered surface and the implied-vol fit: the calibrator recovers the truth to a maximum of 0.6 vol bps in 33 evaluations (1.01 s wall, 28 vertices), with the skew (higher local vol at low strike) faithfully reconstructed.

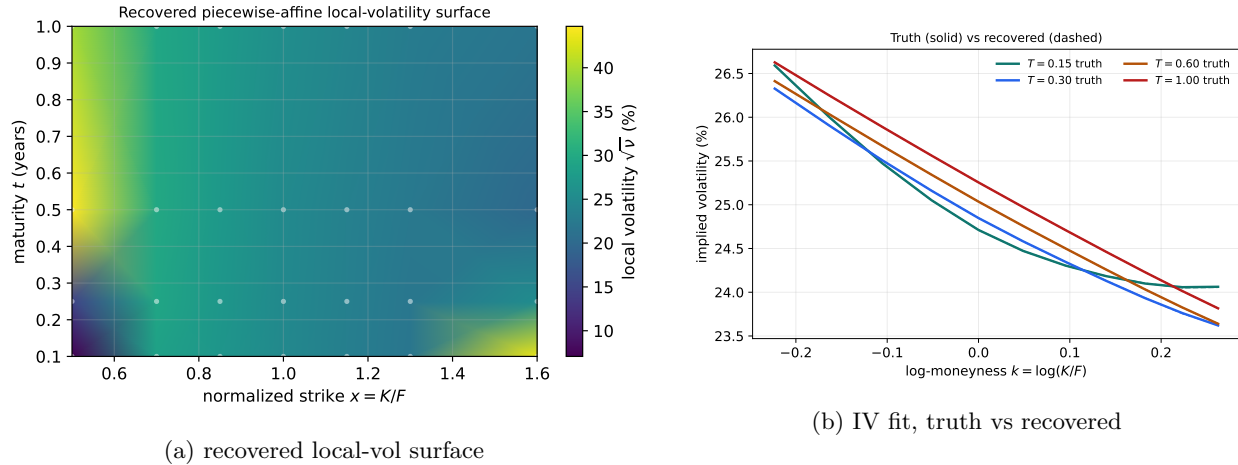


Figure 1: Synthetic round trip through the real calibrator: the skewed surface is recovered to 0.6 vol bps from a flat seed.

## 10.2 The Bloomberg benchmark

On the static Bloomberg fixture the production fit reaches the quality in table 1 and fig. 2: 2.8 vol bps surface RMS on SPY and 11.7 on NVDA, with no node at a bound on the easy expiries. The harder NVDA short-dated expiries carry the larger error — the regime section 9 addresses.

Table 1: Production local-vol fit on the Bloomberg fixture (fresh run).

| Name | surface RMS (bp) | worst expiry (bp) |
|------|------------------|-------------------|
| SPY  | 2.8              | 4.4               |
| NVDA | 11.7             | 12.4              |

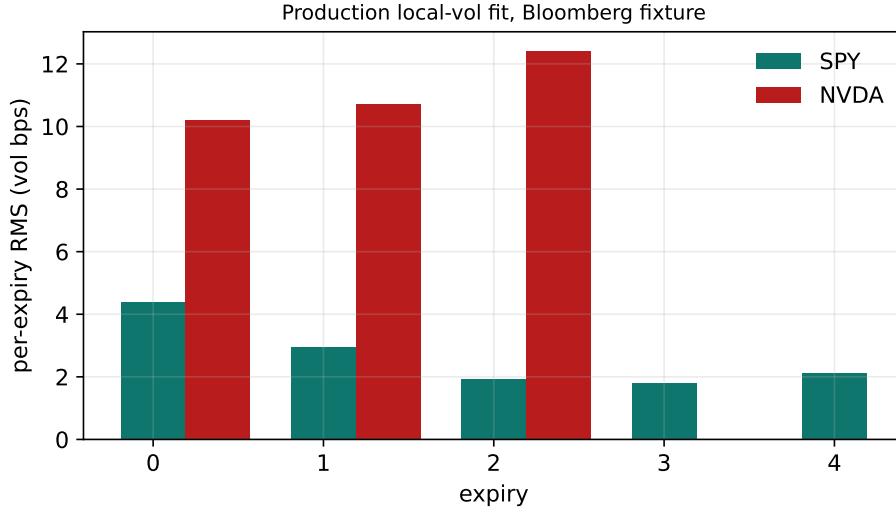


Figure 2: Per-expiry weighted RMS for SPY and NVDA on the fixture.

## 11 What is original, and limitations

The  $P_1$ -positivity guarantee and the forward-calibration philosophy are Andreasen–Huge; the contributions here are engineering depth. The *vectorized-Thomas Numba march* (loop order as the lever,  $6.5\times$ ), the *stall early-stop* that scales the whole fit, the *matrix-free GN* correctly scoped to where the SVD actually dominates, the *source-PDE var-swap* with analytic sensitivities, and the *measure-first* short-dated diagnosis are each non-obvious and were validated against a real benchmark that twice caught regressions a synthetic would have missed. Limitations: the model is a pure diffusion (no jumps), so a true short-dated event spike is only approximated; the tensor grid wastes vertices in the corners (the future non-tensor bowtie grid is where GN’s no-SVD advantage would finally dominate); and the fit, though much faster, remains the heaviest single computation in the app.

## A Hyperparameter atlas

Table 2: Local-volatility hyperparameters.

| Knob                              | Default | Role |
|-----------------------------------|---------|------|
| <i>Surfaced (OptionsSettings)</i> |         |      |

| Knob                                                               | Default                                   | Role                                                           |
|--------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------|
| <code>gridXNodes</code>                                            | 12                                        | Strike vertices of the variance surface.                       |
| <code>gridXMinPerExpiry</code>                                     | 8                                         | Min in-range strike vertices per expiry (section 9).           |
| <code>gridTNodes</code>                                            | 10                                        | Maturity vertices.                                             |
| <code>gridRegLambda <math>\lambda</math></code>                    | $10^{-2}$                                 | Roughness penalty strength.                                    |
| <code>gridRegRho <math>\rho</math></code>                          | 1.0                                       | Time-vs-strike roughness balance.                              |
| <code>convexWingWeight</code>                                      | $10^3$                                    | Convex-wing hinge weight (tail only).                          |
| <code>frontTieWeight</code>                                        | $10^{-2}$                                 | Front-end tie-to-ATM weight.                                   |
| <code>lvVolCapMult</code>                                          | 3.0                                       | Local-vol cap as a multiple of max implied vol.                |
| <code>leftWingSlopeMult</code>                                     | 1.5                                       | Left local-variance extrapolation slope multiple.              |
| <code>varSwapMethod</code>                                         | <code>static</code>                       | Var-swap pricer: replication or source-PDE.                    |
| <code>lvSolver</code>                                              | <code>gn</code>                           | Gauss–Newton (default) or <code>trf</code> .                   |
| <code>lvFastKernel</code>                                          | <code>true</code>                         | Numba vectorized-Thomas march.                                 |
| <code>lvEarlyStop</code>                                           | <code>true</code>                         | Stall-based early stop.                                        |
| <code>timeScheme</code>                                            | <code>implicit</code>                     | Implicit Euler or Rannacher/CN.                                |
| <i>Hidden (affine-calibration PDE / solver)</i>                    |                                           |                                                                |
| affine PDE step                                                    | $\Delta x = 0.01$                         | Spatial step on $x \in [0, \sim 2.5]$ ( $\sim 250$ nodes).     |
| affine time step                                                   | $\Delta T = 0.01$                         | Implicit-Euler maturity step.                                  |
| <code>rannacher_steps</code>                                       | 2                                         | Implicit start-up steps before CN (opt-in scheme).             |
| short-dated step                                                   | $0.3 \sigma \sqrt{\tau}$                  | Refined $\Delta x$ floor, clamped $[1/400, 0.01]$ (section 9). |
| <code>VAR_FLOOR</code>                                             | $10^{-6}$                                 | Dupire local-variance floor.                                   |
| variance bounds                                                    | $[0.005, 0.20]$                           | Nodal local-variance box (vol $[7\%, 45\%]$ ).                 |
| <code>x_scale</code>                                               | <code>jac</code>                          | Optimizer column scaling.                                      |
| <code>xtol/ftol/gtol</code>                                        | $10^{-8}$                                 | Optimizer tolerances.                                          |
| <code>max_nfev</code>                                              | 200                                       | Evaluation cap.                                                |
| <code>stall_window/rtol</code>                                     | 12–18 / $\sim 3\text{--}5 \times 10^{-3}$ | Early-stop window/tolerance.                                   |
| <i>Hidden (separate log-strike validator, <code>pde.py</code>)</i> |                                           |                                                                |
| <code>DEFAULT_N_K</code>                                           | 1201                                      | Validator spatial nodes.                                       |
| <code>DEFAULT_DT_MAX</code>                                        | 1/400                                     | Validator max time step.                                       |
| <code>N_RANNACHER</code>                                           | 4                                         | Validator Rannacher quarter-steps.                             |
| <code>SPAN_SD / SPAN_MIN</code>                                    | 7.5 / 0.6                                 | Validator grid span (sd / floor).                              |

## B Performance notes

1. **Numba march** (`lvFastKernel`). No-pivot factor-once Thomas with the  $m$  sensitivity columns as the contiguous inner SIMD loop and fused source:  $6.1\text{--}6.9\times$  vs LAPACK `dgbstv`, exact to  $\sim 10^{-15}$ . Graceful banded fallback if Numba is absent.
2. **Early stop** (`lvEarlyStop`). Best-iterate stop on option-block stall; scales the whole fit,  $1.45\times$  (SPY) to  $3.3\times$  (NVDA),  $+0.1\text{--}0.25$  vol bp. `stall_window=0` is byte-identical.
3. **Matrix-free GN** (`lvSolver=gn`). Preconditioned LSMR on  $Jv/J^\top w$ ; avoids the dense SVD.

1.3–1.65 $\times$  over TRF at tensor-grid sizes; gated to the smooth mid target.

4. **Warm starts.** Previous-surface seed ( $\sim 38\times$  recal) and parametric Dupire cold seed (1.3–1.8 $\times$ ).
5. **Shelved (documented):** coarse grid (biases  $\theta$ ), first Numba attempt (wrong loop order), Rannacher (1.1 $\times$  + arb risk), GN for the non-smooth band objective, thread/process parallelism (GIL). Four distributed-cost dead-ends confirming the cost is spread, not localized.

## C Reference implementation: Dupire extraction

Forward calibration never differentiates the data (section 2), but the inverse map — read a local-variance grid off an *already-fitted* total-variance surface  $w(k, T)$  — is Gatheral’s formula, used for grid export and the graph LV projection (Note 14). The denominator  $g$  is the butterfly function:  $g \leq 0$  marks strike arbitrage in the implied surface, where no positive local variance can reproduce it, returned `nan`. Matches production to  $10^{-13}$ .

Listing 2: Dupire/Gatheral local variance from an implied total-variance surface (distilled from `models/localvol/dupire.py`).

```
1 def dupire_local_variance(k, w, wk, wkk, wt): # w=w(k,T) with its k,
      kk, T derivatives
2     r = k/w
3     g = 1 - r*wk + 0.25*(-0.25 - 1/w + r*r)*wk*wk + 0.5*wkk #
      butterfly denominator
4     return np.where((w > 0) & (g > 0), wt/np.where(g > 0, g, 1), np.
      nan) # nan if g <= 0
```

## References

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